

Advanced (Habanero)

Q1. Solve the equation $x^2 + 5x + 6 = 0$ by factoring.

- (A) $x = -2, x = -3$
- (B) $x = -1, x = -6$
- (C) $x = 2, x = 3$
- (D) $x = 1, x = 6$
- (E) $x = -2, x = 3$

Q2. Solve the equation $x^2 - 7x + 12 = 0$ by factoring.

- (A) $x = 3, x = 4$
- (B) $x = 2, x = 6$
- (C) $x = 1, x = 12$
- (D) $x = 5, x = 12$
- (E) $x = 3, x = 12$

Q3. Solve the equation $x^2 + 2x - 15 = 0$ by factoring.

- (A) $x = -5, x = 3$
- (B) $x = 5, x = -3$
- (C) $x = 1, x = 15$
- (D) $x = -1, x = -15$
- (E) $x = 2, x = 15$

Q4. Solve the equation $x^2 - 4x - 21 = 0$ by factoring.

- (A) $x = 7, x = -3$
- (B) $x = -7, x = 3$
- (C) $x = 2, x = 21$
- (D) $x = -2, x = -21$
- (E) $x = 1, x = 21$

Q5. Solve the equation $x^2 + 10x + 24 = 0$ by factoring.

- (A) $x = -4, x = -6$
- (B) $x = 4, x = 6$
- (C) $x = 2, x = 12$
- (D) $x = -2, x = -12$
- (E) $x = 1, x = 24$

Q6. Solve the equation $x^2 - 9x + 20 = 0$ by factoring.

- (A) $x = 4, x = 5$
- (B) $x = 2, x = 10$
- (C) $x = 1, x = 20$
- (D) $x = -1, x = -20$
- (E) $x = 9, x = 20$

Q7. Solve the equation $x^2 + 6x + 8 = 0$ by factoring.

- (A) $x = -2, x = -4$
- (B) $x = 2, x = 4$
- (C) $x = 1, x = 8$
- (D) $x = -1, x = -8$
- (E) $x = 3, x = 8$

Q8. Solve the equation $x^2 - 2x - 15 = 0$ by factoring.

- (A) $x = -3, x = 5$
- (B) $x = 3, x = -5$
- (C) $x = 1, x = 15$
- (D) $x = -1, x = -15$
- (E) $x = 2, x = 15$

Open Ended Questions (FRQ)**Q9.** Solve the equation $x^2 + 4x - 8 = 0$ by factoring and verify your solution.**Q10.** Solve the equation $x^2 - 5x - 6 = 0$ by factoring and verify your solution.**Chef's Tips**

- Tip 1: Make sure to factor the left side of the equation.
- Tip 2: Use the zero-product property to solve for x.
- Tip 3: Verify the solutions by plugging them back into the original equation.